MILITARY SPECIFICATION

CYLINDER ASSEMBLY

This specification is approved for use by Picatinny Arsenal, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the inspection and test of the 2 piece cylinder used to contain M56 anti-tank mines.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effort on date of invitation for-bids or request for proposal, forms a Part of this specification to the extent specified herin.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specifications For

STANDARDS

FEDERAL

FED-STD-151 - Federal Test Method Standard, Metals; Test Method

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-1169- Packaging, Packing and Marking For Shipment of Inert Ammunition Components

FSC: 1345

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2.2 Other publications. -The following document forms a part of this specification to the extent specified herein. unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING MD MATERIALS

ASTM Designation E18 - Standard Methods of test for Rockwell Hardness.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Penn. 19103).

DRAWINGS

US ARMY PICATINNY ARSENAL

9312693 - Cylinder Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer) .

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with the applicable drawings and specifications.

3.2 Assembly. - The assembly shall comply with all requirements specified on Drawing (dwg. 9312693 and with all requirements specified on applicable specifications.

3.3 Hydrostatic pressure test

3.3.1 Low pressure. - The cylinder shall not rupture, leak or deform when subjetted to an internal hydrostatic pressure of one hundred plus ten (100 + 10) pounds per square inch gage (psig) for a period of ten plus one (10 + 1) seconds.

3.3.2 High pressure. - The cylinder shall not rupture, leak or deform when subjetted to an internal hydrostatic pressure of three-hundred plus ten (300 + 10) psig for a period of ten plus one (10 + 1) seconds.

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3.4 Mechanical properties. - Cold-working or heat-treating (including quenching-and-tempering, normalizing, normalizing-and-tempering, annealing, and stress-relieving) operations, shall be performed in such a manner as to produce the specified properties uniformly throughout the entire lot. The contractor shall determine the detailed procedures to produce parts meeting the required mechanical properties. These procedures shall be documented and submitted with the first article sample to the Contracting Officer. Changes in these procedures shall also be documented and submitted to the Contracting Officer.

3.5 Workmanship - All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks and surface cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. All required markings shall be neat and sharply defined.

3.6 Submission of results of contractor-conducted examinations and tests. - The contractor shall forward copies of the results required by the contract (See 6.4).

3.7 First article inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provision. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this specification.

4.2 Classification of inspection. - The following type of inspections shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

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4.3.1 Submission. - The contractor shall submit a first article sample (See 6.2B) as designated by the Contracting Officer. The first article sample shall consist of:

a. Ten (10) sets of all components
b. Twenty-five (25) cylinder assemblies

The first article sample shall be evaluated in accordance with the provisions of 4.3.2 and 4.3.3.

*4.3.2 Inspections to be performed. - See MIL-A-48078 and paragraphs 4.4.2.1 and 4.4.2.2, testing per paragraph 4.4.5 and Table I specified herein.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-C-63093 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Cylinder Assembly				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Parts				
	Head Cylinder (See Dwg. 9312692)	10	100%	3.1	4.4.2.1
	Assembly				
	Cylinder Assembly (See Dwg. 9312693)	10	100%	3.2	4.4.2.2
	Testing				
	Hardness	25			4.5.1
	Hydrostatic Low Pressure	25			4.5.2
	Weld Hardness	25			4.5.4
	Hydrostatic High Pressure	20(a)			4.5.3
	Salt Spray	5(a)			4.4.3.1

NOTES

(a) Tests shall be performed in the sequence indicated above.

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4 .3.3 Rejection. - (See MIL-A-48078)

4.4 Quality conformance inspection. -

4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. - (See MIL-A-48078). - Unless otherwise specified inthe classification of defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Equipment necessary for the performance of the inspection listed shall be in accordance with 4.4.4.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63093 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Head Cylinder				9312692
					NEXT HIGHER ASSEMBLY 9312693
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Depth of pressure well		0.40%	3.2	Gage
102	Location of helix angle basic points on lug		0.40%	3.2	Gage
103	Lugs diameter		0.40%	3.2	Gage
104	Lugs diameter location		0.40%	3.2	Gage
105	Propagation hole diameter		0.40%	3.2	Gage
Minor					
201	Number of teeth		0.65%	3.2	Visual
202	Location of propagation hole		0.65%	3.2	Visual
203	Gas chamber diameter		0.65%	3.2	Visual
204	Evidence of poor workmanship		1.0%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63093 (PA)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.2	Cylinder Assembly		9312693
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter of charge threads		100%	3.2	Gage
102	Distance from top of cylinder to inside surface of cylinder containing propagation hole		0.40%	3.2	Gage
103	Minimum length of charge thread		0.40%	3.2	Gage
104	Overall outside length from bottom opening to shoulder		0.40%	3.2	Gage
105	Outside diameter (in restrained condition in accordance with the drawing)		0.40%	3.2	Gage
106	Location of top diameter		0.40%	3.2	Gage
107	Perpendicularity of cylinder head to tube		0.40%	3.2	Gage
108	Distance from inside of cylinder at minimum thickness to top surface of cylinder large diameter		0.40%	3.2	Gage
109	Inside diameter of large diameter (in restrained condition and in accordance with the drawing)		0.40%	3.2	Gage
110	Welds extending past maximum of outside diameter		0.40%	3.2	Gage
111	Cartridge thread location		0.40%	3.2	Gage

NOTES

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63093 (PA)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.2	Cylinder Assembly				9312693
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
Minor					
201	Burrs or high spots on chamfers inside large diameter		0.65%	3.2	Gage
202	Location of wrench slots		0.65%	3.2	Gage
203	Overall length		0.65%	3.2	Gage
204	Evidence of poor workmanship		1.0%	3.5	Visual

NOTES:

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4.4.3 Testing

4. 4.3.1 Salt spray (See dwg. 9312693) Major Defect - Four (4) cylinders shall be randomly selected from each lot and subjected to the salt spray test requirements specified on the applicable drawings. Testing shall be in accordance with paragraph 811.1 of FED-STD-151. Any cylinder failing to comply with all of the specified requirements shall be classed defective. The lot shall be rejected if one (1) or more cylinders are classed defective. Testing is destructive and samples shall not be returned to the lot.

4.4.3.2 Hardness test (See dug. 9312693) Major Defect - Three hundred and fifteen (315) clyinders shall be randomly selected from each lot and tested in accordacne with 4.5.1. The lot shall be rejected if three (3) or more cylinders are classed defective.

4.4.3.3 Hydrostatic Low Pressure Test. - (see 3.3.1) Major Defect - Three hundred and fifteen (315) cylinders shall be-tested in accordance with 4.5.2. The lot shall be rejected if one (1) or more cylinders are classed defective (Non-destructive test).

4.4.3.4 Hydrostatic high pressure test. - (See 3.3.2), Major Defect - Twenty-five 25) cylinders that have successfully passed they hydrostatic low pressure test of 4.4.3.3 shall be tested in accordance with 4.5.3. The lot shall be rejected if one (1) or more cylinders are classed defective (Destructive test).

4.4.3.5 Weld hardness test (See dwg. 9312693) Major Defect - Three hundred and fifteen (315) cylinders shall be randomly selected from each lot and tested in accordance with 4.5.4. The lot shall be rejected if *three* (3) or more cylinders are classed defective (Non-destructive test) .

4.4.4 Inspection equipment. - The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the 'Paragraph Reference/ Inspection Method' column in the tables starting with paragraph 4.4.2.1, and the test method paragraphs (see 4.5). The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078, and 6.2 herein.

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